

GULFAM HASSAN

Senior Software Engineer | Backend, Distributed Systems & Cloud Infrastructure | Agentic AI & Automation

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SUMMARY

Senior Backend Engineer with **9+ years** of experience building and scaling **distributed systems** that serve **millions of users**, primarily in **Java, Spring Boot, Kafka, and Kubernetes on AWS**. Built **real-time, event-driven systems** processing **10M+ records a day**, reduced **p99 API latency by 45%**, and owned **production reliability** through **on-call, observability, and incident response**. Experienced in leading **system design (HLD/LLD)** end to end and establishing **SLO-driven engineering** practices across teams. On the AI side, built an **agentic RCA workflow using LangGraph and LLMs** that automates **incident root-cause analysis** in production - collecting and aggregating context across ELK, Kafka, Redis, and Kubernetes - with hands-on experience in **agent orchestration, tool integration, and prompt engineering**.

TECHNICAL SKILLS

Languages: Java (8-21), Python, TypeScript

Backend & APIs: Spring Boot, Spring Cloud, Spring Batch, Hibernate, JPA, REST, WebSocket, gRPC, Microservices, Multithreading, Concurrency

System Design: HLD, LLD, Scalability, CAP Theorem, Sharding, Partitioning, Replication, Load Balancing, Rate Limiting, Circuit Breakers, Design Patterns

Distributed Systems: Apache Kafka, Event-Driven Architecture, CQRS, Saga Pattern, Idempotency, Fault Tolerance, Async Messaging, Distributed Caching

Databases & Caching: MySQL, PostgreSQL, Oracle, Redis, Elasticsearch

Cloud & DevOps: AWS (EC2, ECS, EKS, Lambda, RDS, S3, SQS, SNS, CloudWatch, IAM), Docker, Kubernetes, Jenkins, CI/CD, Git, Terraform

Observability & SRE: ELK Stack, Prometheus, Grafana, Splunk, SLI/SLO/SLA, On-call, Incident Response, Distributed Tracing (Zipkin), Domain-Driven Design

Testing & Security: JUnit, Mockito, TestNG, SonarQube, Snyk, TDD, Integration Testing

AI & Agentic Systems: LLM Integration, LangGraph, RAG, Agentic Workflows, Agent Orchestration, Prompt Engineering, LLM-based RCA/Automation

Practices: Agile, Scrum, Code Reviews, Design Docs, Mentoring, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Amex GBT | Senior Technical Lead | May 2024 - Present

- Led the **HLD and LLD** for a **multi-tenant backend platform** serving **millions of users**, designing it to scale horizontally across **AWS** regions and hold **99.99% availability** under production load.
- Built **Spring Batch** pipelines that process **10M+ records a day** using dynamic partitioning and exponential-backoff retries for fault tolerance. Cut **batch runtime by 40%** and eliminated the recurring SLA breaches.
- Brought **p99 API latency down by 45%** by adding **Redis caching**, moving hot paths to **CompletableFuture** for async processing, and routing reads to **DB replicas**.
- Containerized and deployed **microservices** using **Docker and Kubernetes (EKS)** and rebuilt our **Jenkins CI/CD** pipelines so teams could release daily instead of weekly, improving **deployment success rate by 25%**.
- Set up monitoring and alerting with **ELK, Prometheus, and Grafana** - cut **incident detection time by 40%**. Right-sizing instances and tuning autoscaling on top of that took **15-25% off our AWS bill**.
- Added **Snyk** to our CI/CD for code analysis and dependency scanning and pushed **unit test coverage from 60% to 85%+** with **JUnit, Mockito, and SonarQube**. Cut **high-severity vulnerabilities by 70%** and dropped post-release production defects from roughly **8-10 per release to 2-3**.

- Built an **agentic AI RCA workflow** using **LangGraph and LLMs** to automate incident investigation for failed transactions - a **Python** listener triggers on production failures (provider timeouts, payment errors, high-CPU events), collects and aggregates context from **ELK, Kafka, Redis, Kubernetes, and databases**, and generates a **root-cause analysis** routed to **Slack, email, and Jira**, cutting manual investigation time during incidents.

Altimetrik | Senior Product & Platform Engineer | May 2021 - April 2024

- Built **event-driven services** on **Apache Kafka** handling **3,000+ TPS** with **exactly-once semantics**, reducing **end-to-end latency by 30%**.
- Designed the backend services and **orchestration layer** for our **high-throughput data pipelines**, adding **consumer-group scaling, backpressure, and idempotent consumers** to keep the system stable under peak load.
- Part of the **24/7 Black Friday war room** rotation - debugged backend issues live, ran **root-cause analysis** on service failures, coordinated with support during metric spikes, and presented **RCAs** to senior leadership with follow-up actions.
- Built an **automation script** that captured live dashboard snapshots and sent them to leadership and the team on a schedule, removing the need for manual monitoring during peak traffic events.
- Worked with product and SRE to set **SLIs, SLOs, and error budgets** for new services - catching regressions earlier and reducing **customer-impacting incidents by 35%**.
- Pushed **test coverage from 60% to 90%+** with **JUnit, Mockito, and integration tests**, and wrote **design docs** for our core services that became the template other teams adopted.

IBM | Senior Software Engineer | May 2020 - May 2021

- Worked on migrating a legacy **Certificate Information System (CIS → CIS2)** for a global automotive client, rebuilding backend **microservices in Java and Spring Boot** to manage vehicle certification data - specifications, dimensions, and type approvals checked against **country-specific regulatory requirements** before vehicle launch.
- Built **Kafka-based event flows** to sync certification data across services during and after the migration, and improved **data-processing performance by 25%** through efficient query design, index optimization, and connection-pool sizing - verified with **JMeter** load tests before rollout.
- Developed **Angular UI modules** for managing and viewing vehicle certification data, reducing **manual operational effort by 20%**.

Everforth Quinnox | Software Engineer | Nov 2016 - April 2020

- Developed **backend features and REST APIs in Java and Spring Boot** for enterprise client applications, working across the full development cycle from coding to testing to production deployment.
- Built application modules with **Spring Boot, Hibernate/JPA, and Oracle DB**, writing clean, maintainable code and applying **design patterns** (Strategy, Factory, Observer) learned on the job.
- Wrote unit and integration tests with **JUnit**, raising **automated test coverage to 80%** and reducing post-release defects.

EDUCATION

B.Tech, Computer Science Engineering - Guru Gobind Singh Indraprastha University, Delhi (2016)

CERTIFICATIONS & AWARDS

- **Oracle Certified Professional, Java SE 8 Programmer**
- **Performance Excellence Awards** - Altimetrik (2023) and IBM (2021) for platform and reliability impact.